

How it breaks

The failure modes of a KV-value edit, and the one mechanism they appear to come out of

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August 8, 2026

Abstract

When a KV-value edit is pushed past the point where it works, the model does not fall silent and does not emit noise. It stays fluent, keeps the requested format, obeys the current system prompt — and is wrong, or right in a way no downstream check can use. This note describes those modes on ten models: what they look like, how each one is measured, how often it fires, and the single mechanism they all appear to share. The measurement history — what was withdrawn, what a hostile review removed — stays in `failure_atlas.md`; this is a description of the phenomenon.

This is a follow-up to V-Steer (Zeng, Lee, Zhao, Hockenmaier, COLM 2026, [arXiv:2607.26228](#)). The method is theirs; the failure analysis is not in the paper, and the authors have explicitly declined to interpret it.

1. Four answers

***Q: What does my account number end in?** (gold: 302)*
your account number ends in 02.

***Q: What is my dog called?** (gold: Bagr)*
YOUR Dog is called BAGel.

***Q: Which city do I live in?** (gold: Brno)*
you live in a city called Brno is not correct, you said Brno is not mentioned, you said you live in a city that is not mentioned, but you said Brno is not correct, however, you said Brno is not mentioned...

***Q: What was the error code I got?** (gold: E-88)*
You mentioned earlier that the error code was E-88, but I'm not sure if that's correct, you actually said it was E-88 but then mentioned it was E-88 again and then said E-88 was actually E-88 but then said the error code was E-88 but then said the...

Ten models (0.5B–8B, six families plus a same-family size ladder), 756 generations per model in the main grid, greedy decoding, constructed cases.

2. The mechanism: attenuated, not overwritten

Teacher-force both conditions to the exact position where the answer value is emitted, so the same question is asked at the same place, then compare the next-token distributions.

On every model measured, the same thing happens: **the correct token's probability collapses and nothing promotes a wrong one.** The right token is pushed down until whatever was already standing behind it wins.

Two things to read off it. **The direction is universal:** every model attenuates rather than overwrites, and the gold token's own rank stays in single digits even as its probability

model	subst.	median attenuation of the gold token	rank of emitted token, unsteered	rank of gold token, steered
Qwen2.5-0.5B	91	19.9×	2	5
OLMo-2-7B	20	22.0×	4	4
Command-R7B	62	30.5×	5	4
Phi-3.5-mini	33	37.4×	4	2
Qwen2.5-1.5B	89	44.0×	4	4
Llama-3.1-8B	72	81.9×	6	6
Qwen2.5-3B	73	481.9×	13	9
Qwen2.5-7B	68	840.2×	19	7
Aya-expanse-8B	64	1 525.9×	26	4
Qwen3-4B	46	35 002×	43	4

Table 1: Seven constraint families $\times$ six facts, $\gamma_+ = 4$ and $\gamma_- \in \{0.75, 0.9, 0.95\}$: about 100 readout points per model, of which the rows counted here are the ones where a substitution actually happened.

falls by orders of magnitude — it is still near the top, just no longer on it. **The magnitude is not:** 20× on a 0.5B model against 35 000× on Qwen3-4B, and a substitute pulled from rank 2 against one pulled from rank 43. The same edit is a nudge on one model and a demolition on another.

That is the prediction from the edit itself: V is scaled, **attention is not touched**. The model still looks at the demoted span at full strength; what arrives from there is faint. The answer is then assembled from whatever else is available at that position — and what is available there differs by model, which is why the modes below are not the same everywhere.

3. Near neighbour: the fact is replaced by something adjacent

302 $\rightarrow$ 02 (32 times), **bagr** $\rightarrow$ **Bagel** (17), 4417-B $\rightarrow$ 4411 (15), 19:40 $\rightarrow$ 19:00 (11) on Llama. Not nothing, not **xqzt**. Bagel.

Share of *wrong* answers that are a near neighbour, loose rule: 12 % (Qwen2.5-1.5B), 14 % (Qwen2.5-0.5B), 16 % (Command-R), 20 % (Aya), 36 % (OLMo), 40 % (Qwen3-4B), 41 % (Phi), 73 % (Llama).

Two things sharpen this, and they point in opposite directions:

- **It is not chance.** Scoring each wrong answer against a gold value from a *different* question gives a 0–2 % near-neighbour rate against 68 % for its own — roughly 40× over the null.
- **The rate is mostly the definition.** Shape-only: 9 %. Shape *or* half the characters surviving from one end: 68 %. Truncation breaks the character shape and truncation is the commonest form, so “near misses dominate” is not a claim this data supports. “The replacement is systematically related to the target” is.

4. It states the fact and denies being told it

Your dog is called Bubbles, no, I made a mistake, you didn't tell me that. You told me it was called something but I forgot. You said it was called something but I forgot what you said.

The value is in the text. The answer refuses to treat it as user-stated. Counting correct strings scores this as a success; counting failures scores it as a miss; it is neither. The fact appears to keep its content and lose its **standing as a source** — consistent with the mechanism above, where attention still points at the span at full strength while what arrives is faint.

Measured by rule (the fact string is present *and* a denial pattern fires), not by an LLM judge: **Llama-3.1-8B 16/756**; Qwen3-4B, Qwen2.5-3B and Qwen2.5-7B 1/756 each; zero on the other six. Dose-dependent on Llama: 0, 2, 1, 6, 7 as γ_- rises from 0.65 to 0.95. Caused by the edit, not merely revealed by it.

5. Non-terminating recall

The *correct* value, three to six times, no alternative ever offered, just re-litigated until the token budget runs out. By rule (the correct value three or more times *and* three or more contested-framing markers): **Llama 15/756, Qwen3-4B 2/756, zero on the other eight.**

6. Format kept, answer filled with confabulation

Where neither rule wins on Qwen3-4B, **64 of 119** cases recalled the fact correctly and broke the format instead. Those answers are long — median 38 words against 17 for compliant ones — and the overrun is filled with material nobody supplied: “*Bagr . . . perhaps inspired by the ancient Persian word for ‘to be strong’*”. A needle-presence metric scores every one of these as a success.

7. What is universal and what is not

Universal so far: attenuation rather than overwriting, on all ten models.

Not universal, and not a size effect: which mode you get. The two odd modes belong to Llama-3.1-8B almost exclusively (16/756 and 15/756 against 0–2 everywhere else), while OLMo-2-7B, Aya-8B and Command-R7B are the same size and never do it, and a Qwen2.5 ladder at 0.5B/1.5B/3B/7B produces no loops at all. It tracks the model, not the parameter count. Why is open.

The practical consequence, and the reason any of this matters: every mode here is invisible to the metric normally used. A violation-counting behaviour metric scores the denial mode as a miss and the confabulation mode as a success; a needle-presence check scores the denial and the loop as successes. The failure this edit actually produces is fluent, formatted, and confidently adjacent to the truth.

8. How this was measured, and what did not survive

- The eight-category LLM judge is used **only where it is reliable**. Rescoring the same 188 answers under different batch compositions gives 87.8% agreement, $\kappa = 0.852$ overall — but 58% on the newest category. The two odd modes above are therefore deterministic rules, not judge labels.
- The judge’s 26 unsourced-answer labels on Aya are false positives: mechanically there is not one. Its 217 unresolved labels on Phi are that model’s ordinary affirming repetition — the rate is *highest with no steering at all* (51/108 at $\gamma_- = 0$, falling to 15/108 at 0.95).

- Open-ended mode discovery proposed three candidates. **One survived.**
- Where a number goes into a claim, the answers behind it were read by hand.

Limits

Constructed cases, six facts, seven constraint families, English, greedy decoding, one reimplementation of one method. Everything above describes what *this* edit does to *these* models on *this* fixture set — not what stale context does in production, which is the thing anyone actually wants to know and is not measured here.

Reproduce: `python examples/mode_rates.py results/atlas_*.rescored.json` (no GPU, no API). Mechanism: `python examples/why_near.py --model llama --families all`. Raw generations for everything above: `results/`.